

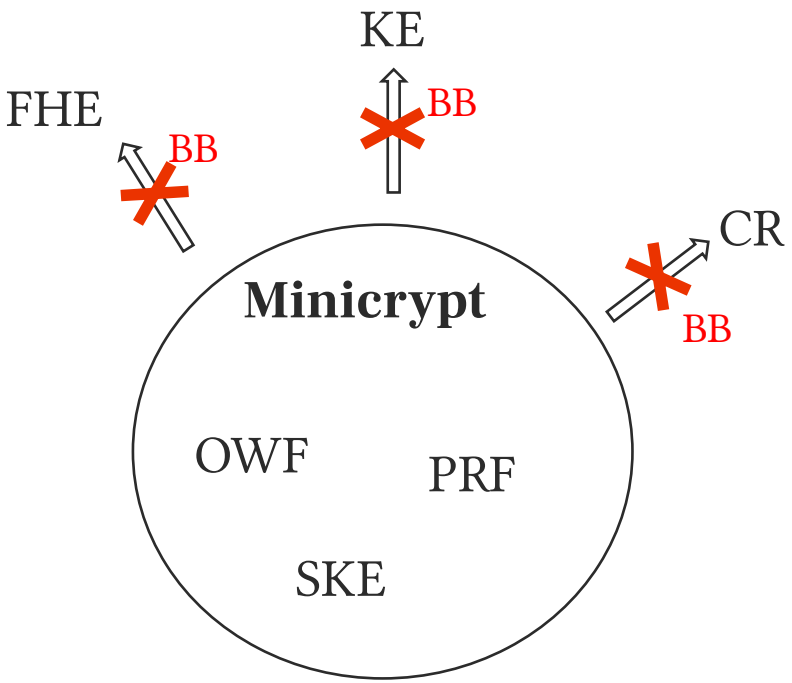
On Composing Oracle Separations

Willy Quach, *Felix Rohrbach*

TCC 2026

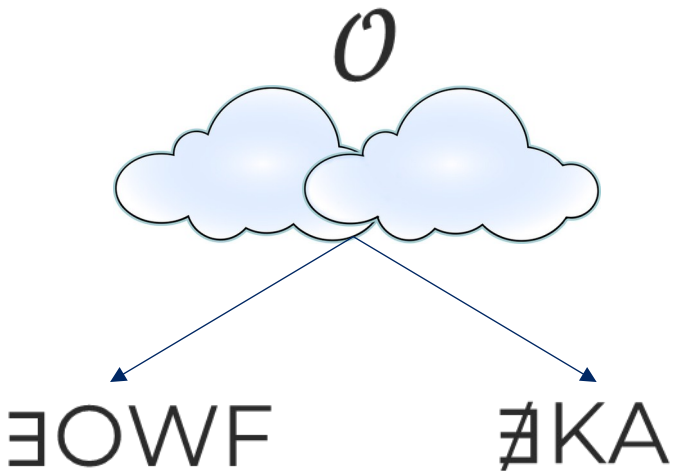
11.11.2026

Minimal Assumptions

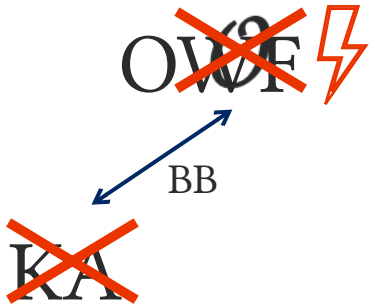


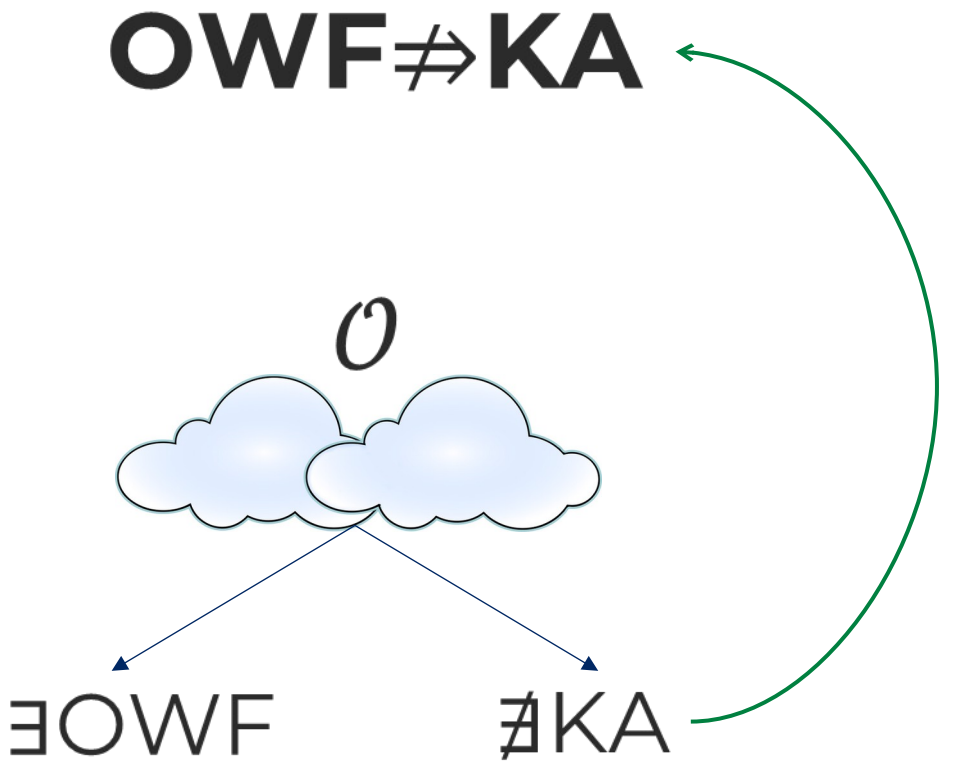
Oracle Separations

$\mathsf{OWF} \not\Rightarrow \mathsf{KA}$



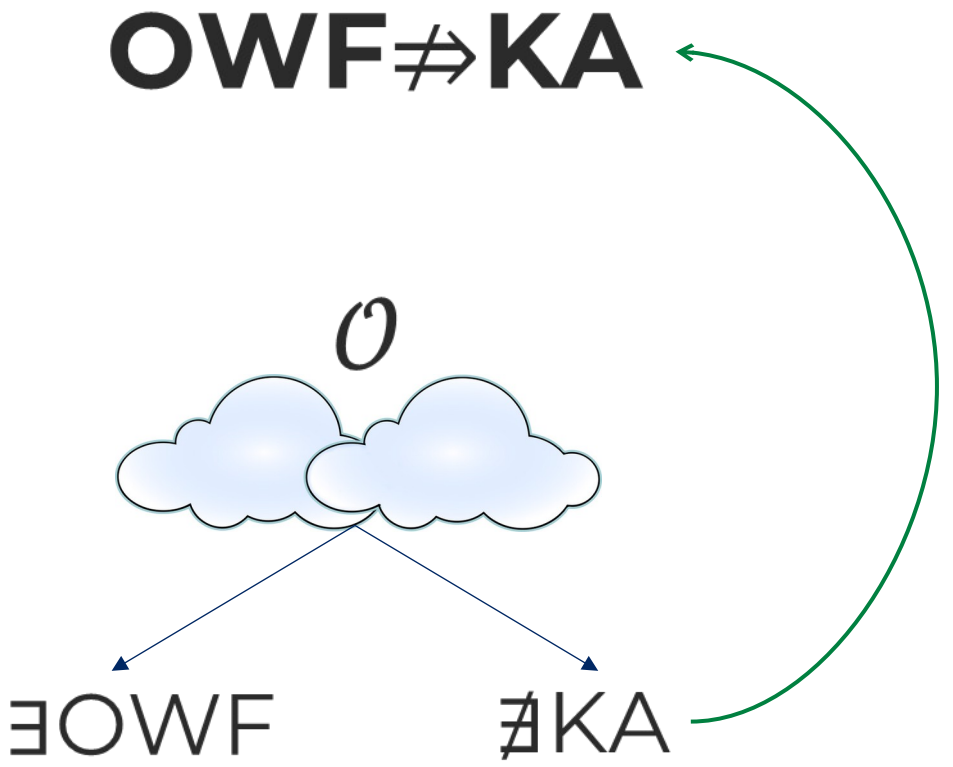
Oracle Separations



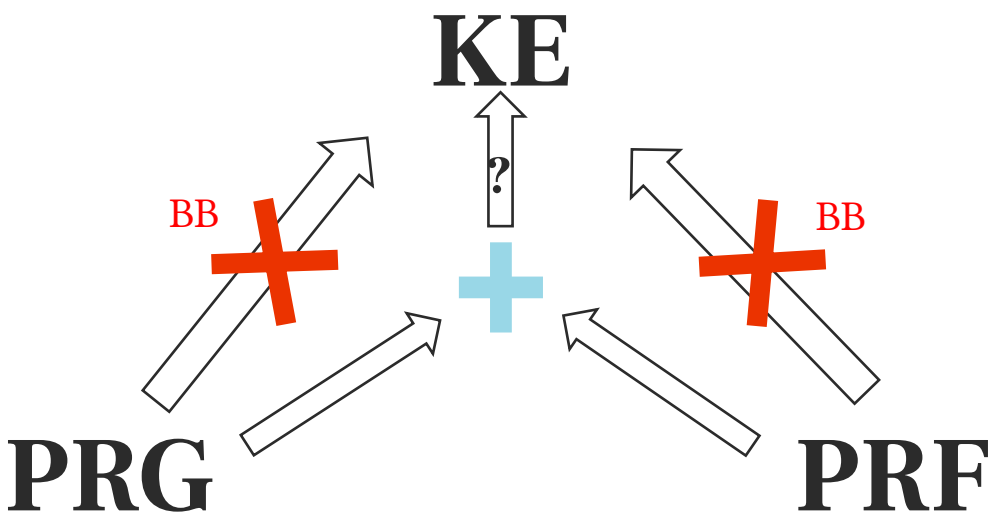


Oracle Separations

- OWF ⇏ Key Agreement [IR'89]
- OWF ⇏ Collision-Resistant Hash Functions [Simon'98]
- PKE ⇎ Oblivious Transfer [GKMRV'00]
- OWF ⇏ Constant-round statistically-hiding commitments [HHRS'07]



What About Composition?





Main Question:
When do Oracle Separations Compose?



When do Oracle Separations Compose?

Black-Box Uselessness:
Composing Separations in Cryptography
Couteau, Farshim, Mahmoody, ITCS'21

- P is not useful to build Q
  - Even in combination with other primitives
  - Formally: If $P + Z \Rightarrow Q$, then $Z \Rightarrow Q$
- Very strong notion



When do Oracle Separations Compose?

Ad-hoc Proofs

- iO + OWP ⇏ CRH [AS'15]
- iO + OWP ⇏ SZK [BDV'17]

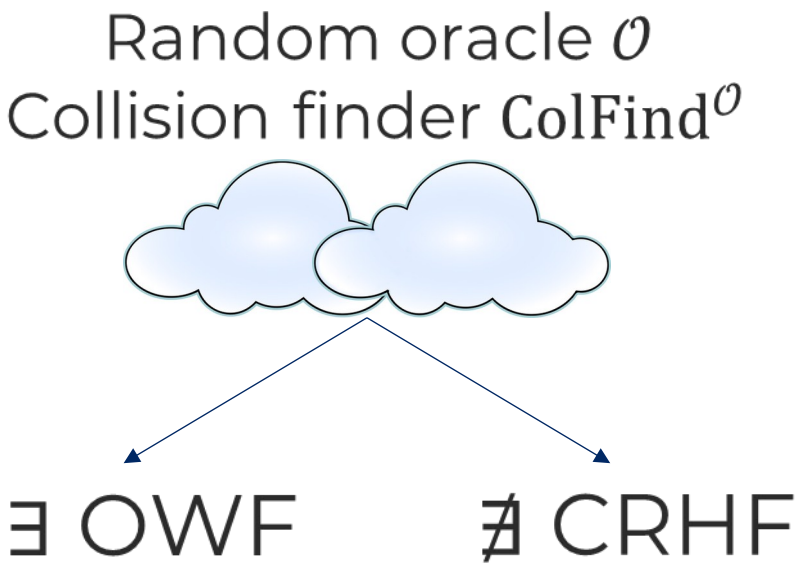
Our Contribution

More general

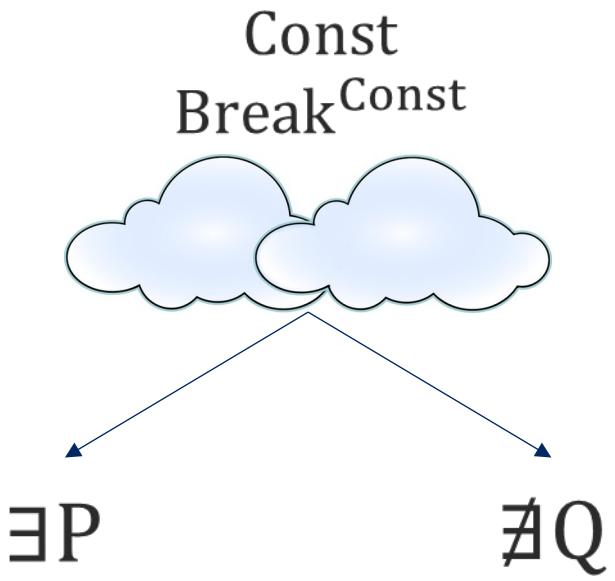
- Ad-hoc Proofs
- **Oracle-composing Separations**
  Natural classes of oracle separations compose
- Black-Box Uselessness

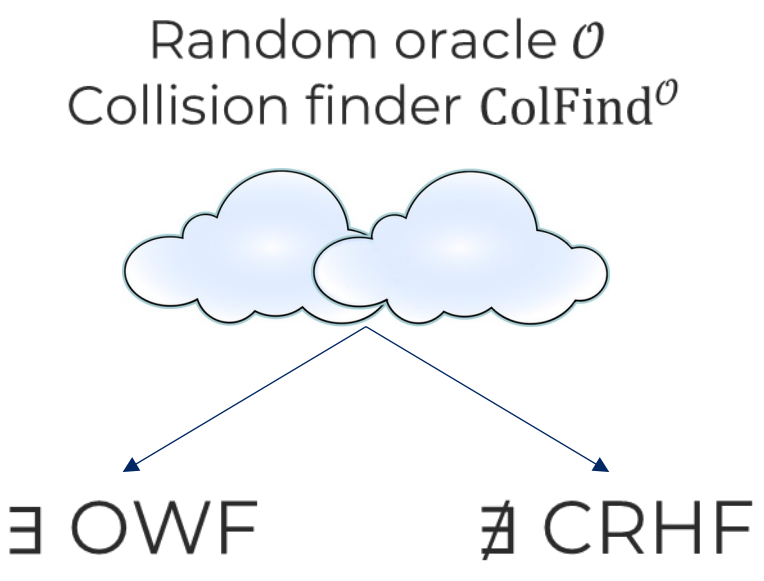
Simon‘s Separation [EC‘98]

- One-way functions $\not\Rightarrow$ Collision-resistant hash functions

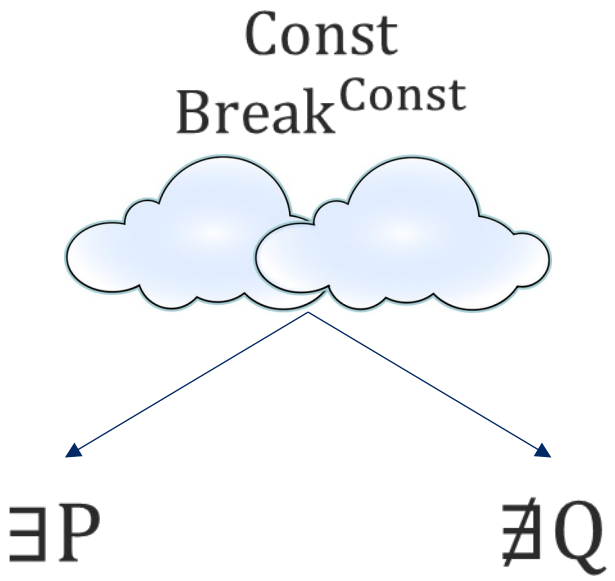


Two-Oracle Separations





Composing Oracle Separations



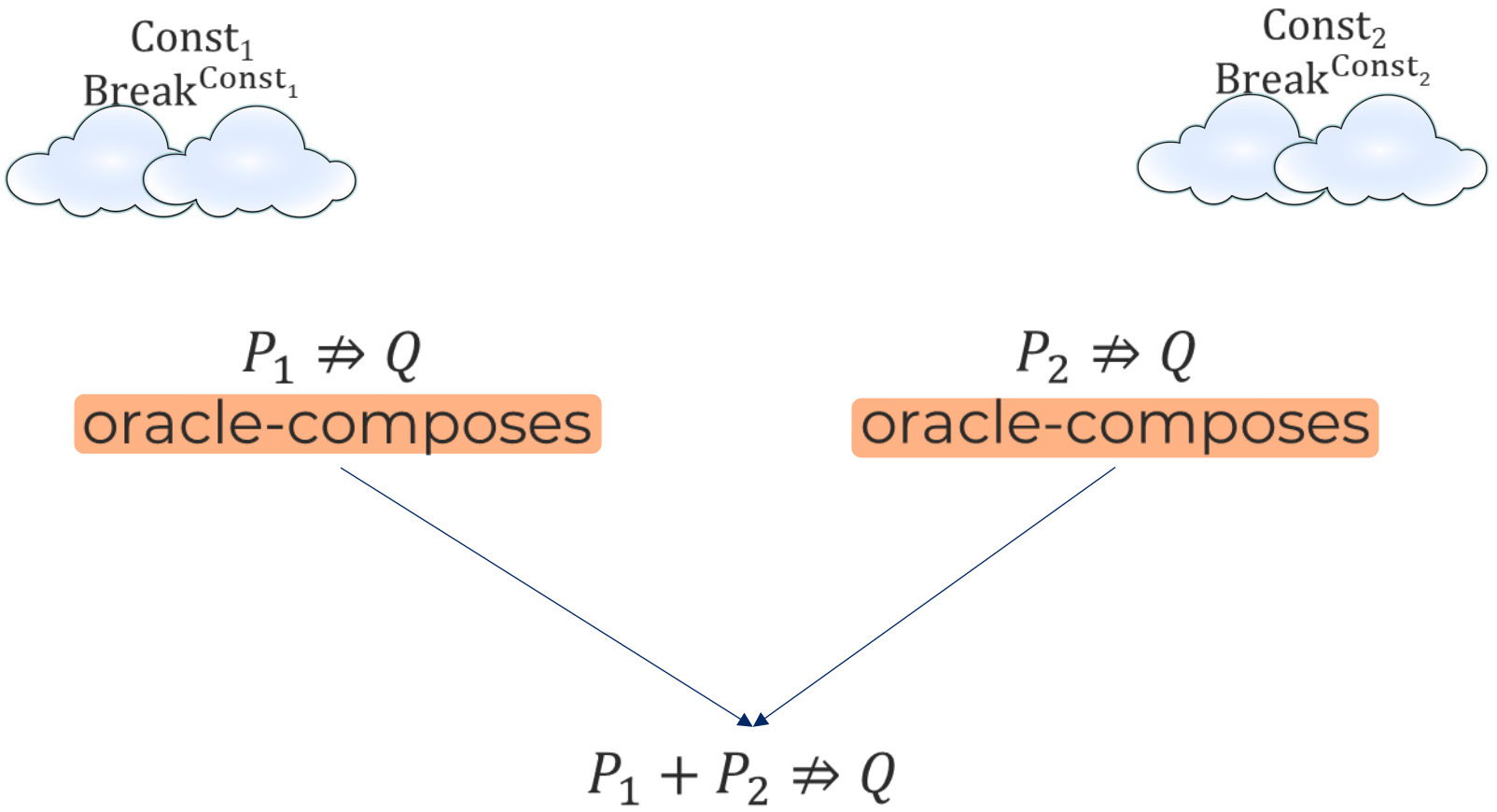
Oracle-separation composes if:

- Break breaks relativizingly:

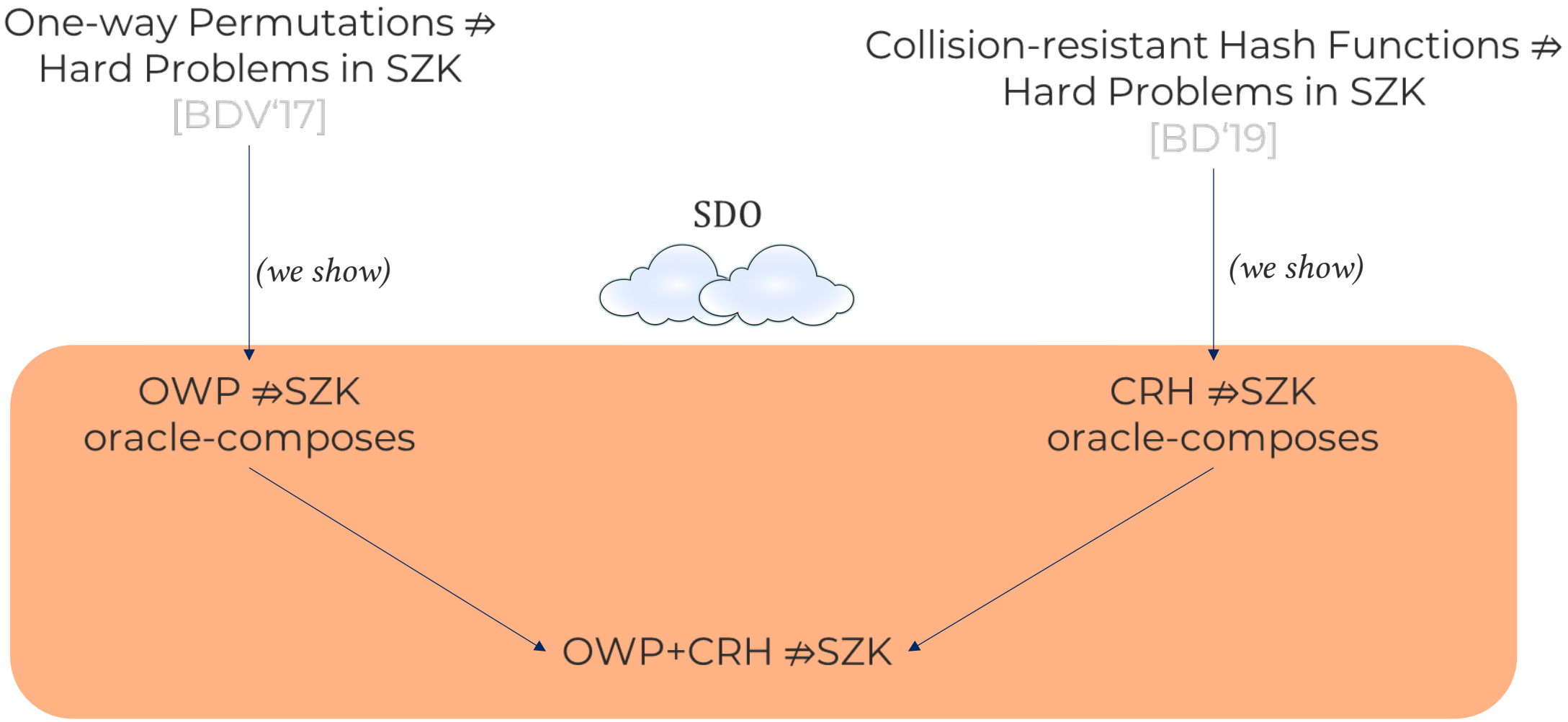
  $\forall Z$: No Q rel. to Z, Break^Z

- Simulation property for Const

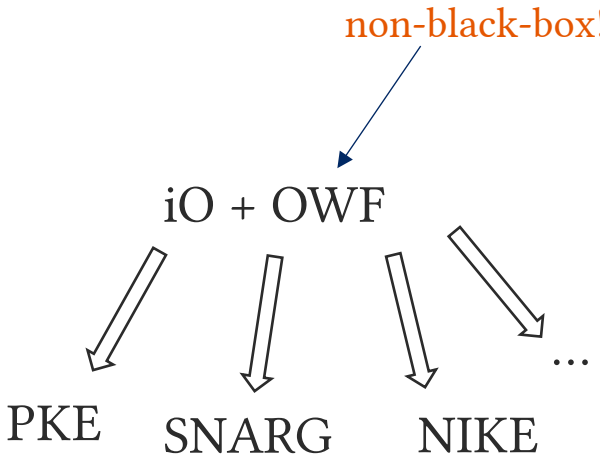
Main Theorem



First Application



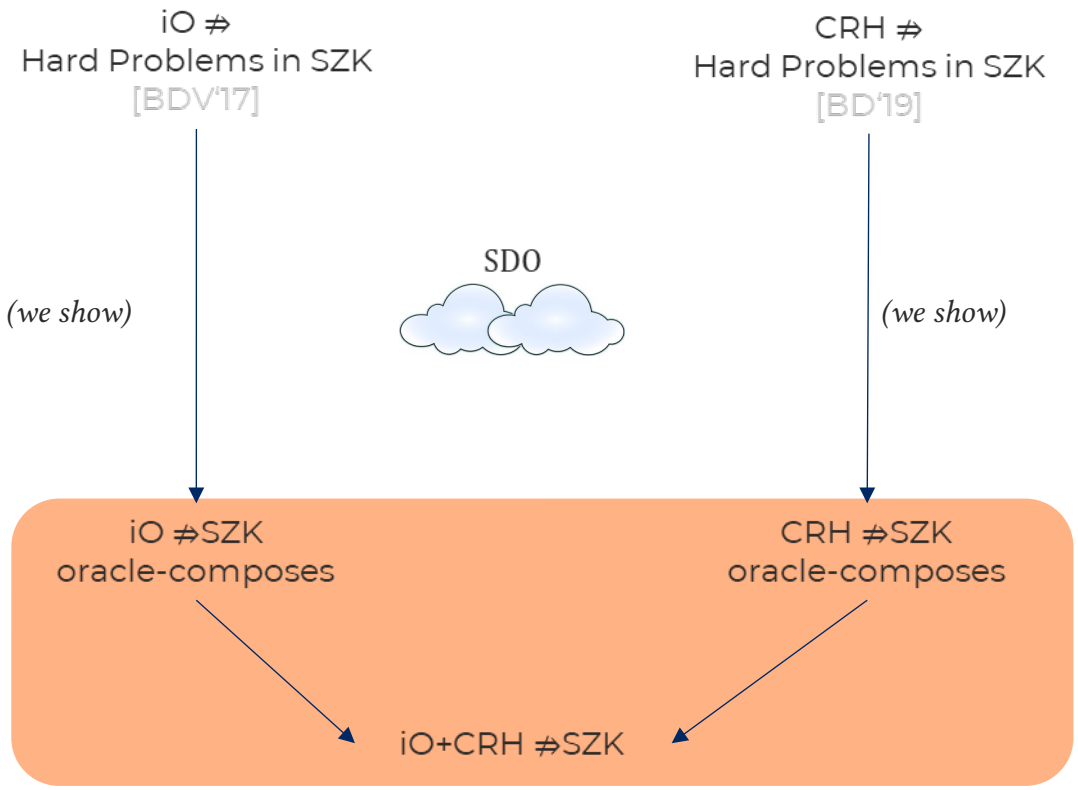
The Case of Indistinguishability Obfuscation



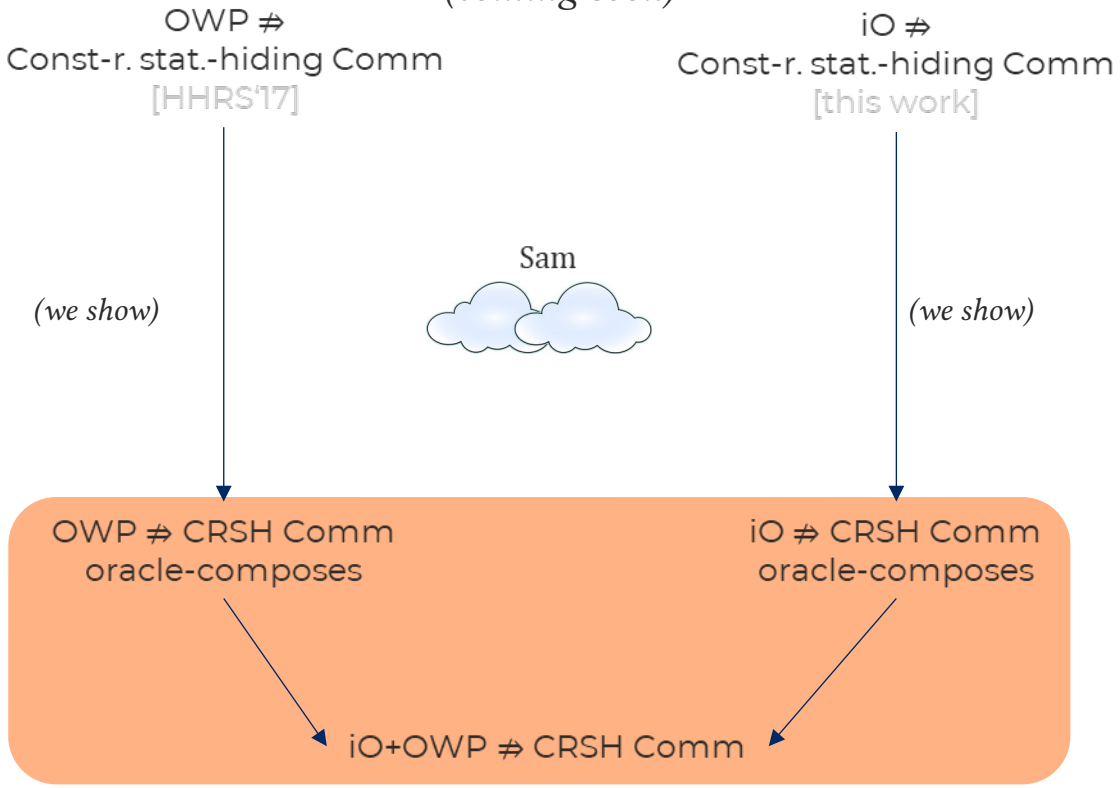
Asharov-Segev [AS'15]:

$$iO^{OWF}(h^{OWF}) \dashrightarrow iO^{Const}(h^{Const})$$

Applications to Indistinguishability Obfuscation

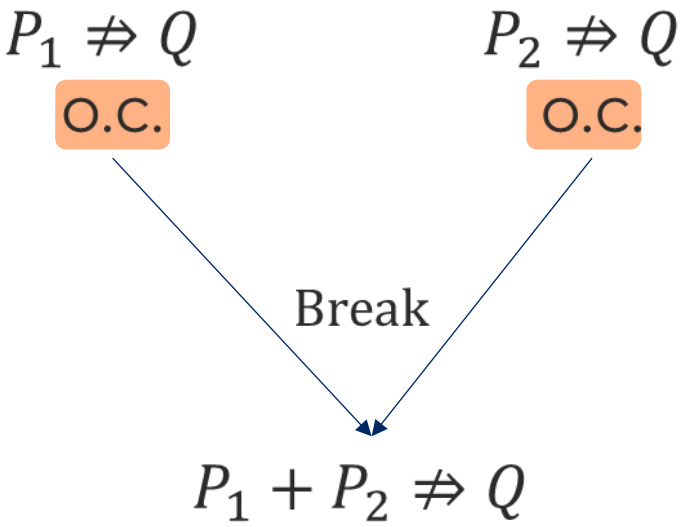


(coming soon)

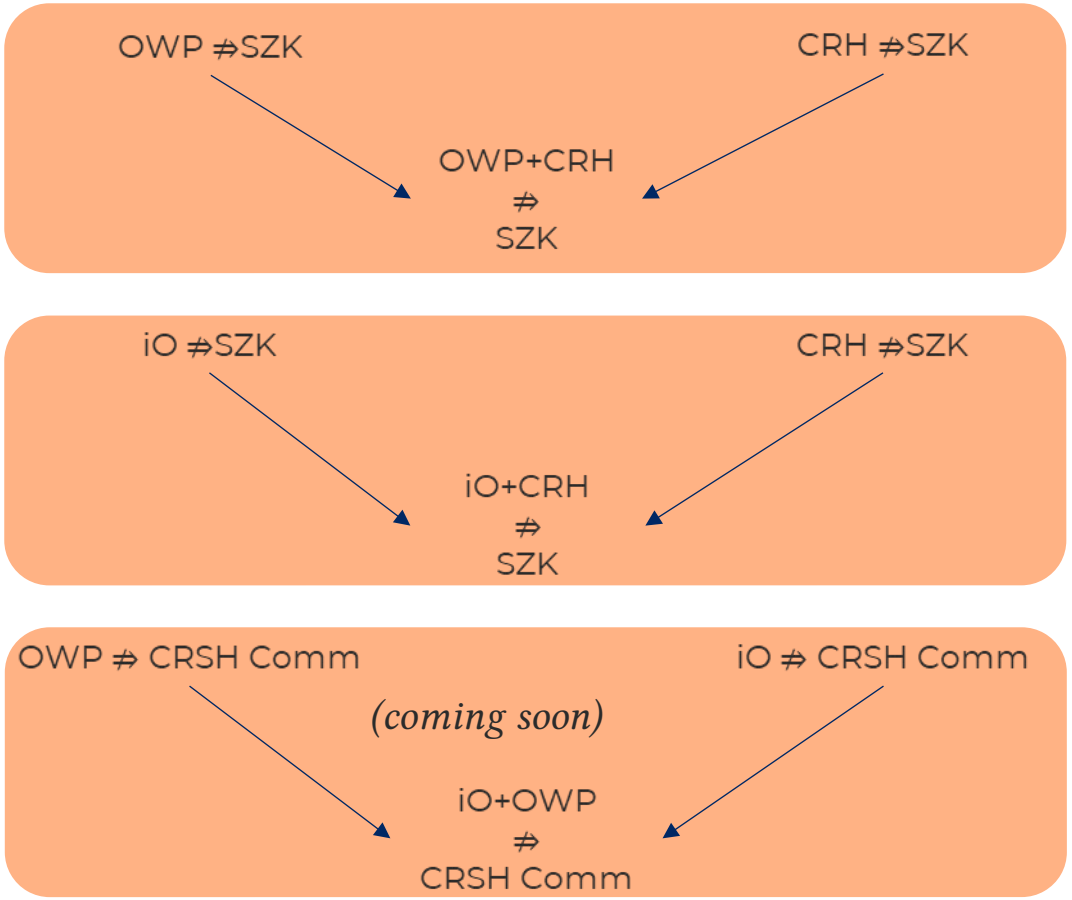


Conclusion

Oracle Composition



Applications



Thank you!